ELE 202 Mid Term W2022 Version 1

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ELE 202 Mid Term ...

Ryerson University
Department of Electrical and Computer Engineering
ELE202-Electric Circuits Analysis
Mid-term Examination
March 4, 2022 - Duration: 120 Minutes
Examiners: Dr. Mohammadi and Dr. Jassar

I solemnly declare that I will complete this examination independently, with full compliance to the Senate Policy 60 Academic Integrity, and meeting the requirements stated in the examination.

Student Last Name: ..... **Student First Name:** .....

Student Number: ..... **Signature:** ..... **(Required)**

NOTES:

The test is Open Book / Open Notes.

There are 5 questions, each with subsections. Answer all questions.

NO QUESTIONS to be asked during the test. If doubt exists as to the interpretation of any question, you are urged to submit with the answer, a clear statement of any logical assumptions made.

Question No.	*Mark of each question*	*Mark obtained*
Q 1	20	
Q 2	20	
Q 3	20	
Q 4	20	
Q 5	20	
Total (100)		

Q 1 (a): [10 marks] The Current versus time graph is given in Figure 1.a. Sketch the charge versus time graph on the current versus time graph. Show all the working steps.

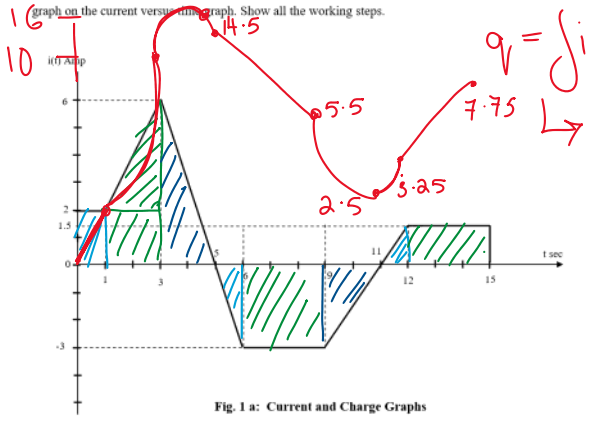


Fig. 1 a: Current and Charge Graphs

$q = \int i\,dt + q_0$

↳ Area

+ve — ⇒

-ve — ⇒

⇒ ∪

⇒ ∩

$t=1 \Rightarrow q = (2\times1) + 0 = 2\,[C]$

$t=3 \Rightarrow q = (2\times2) + \left(\frac{2\times4}{2}\right) + 2 = 10\,[C]$

$t=5 \Rightarrow q = \left(\frac{2\times6}{2}\right) + 10 = 16\,[C]$

$t=6 \Rightarrow q = \left(\frac{1\times-3}{2}\right) + 16 = 14.5\,[C]$

$t=9 \Rightarrow q = (3\times-3) + 14.5 = 5.5\,[C]$

$t=11 \Rightarrow q = (2\times-3) + 5.5 = 2.5\,[C]$

$t=11 \Rightarrow q = \left(\frac{2\times 3}{2}\right) + 5\cdot 5 = 2.5\,[C]$

$t=12 \Rightarrow q = \left(\frac{1\times 1.5}{2}\right) + 2.5 = 3.25\,[C]$

$t=15 \Rightarrow q = (3\times 1.5) + 3.25 = 7.75\,[C]$

Q 1 (b): [10 marks] For the circuit shown in the Fig.1 b below. Fill the table below:

Note: The # inside the red circle refers to the table's component numbers. Assume the polarity of the voltage and direction of current as needed (some of the components are not given with the polarity of the voltage like 2, 3, 5, 8 and the direction of current like component 1).

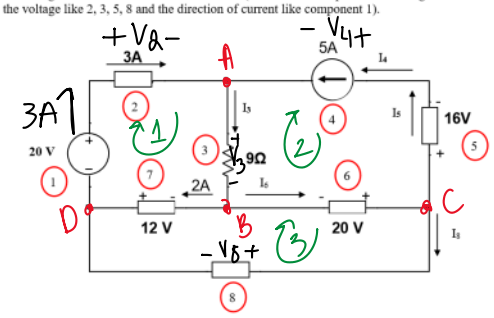


Fig. 1 b: Q1-b

Component Number	Voltage	Current	Power	Power (Absorbed or Supplied)
1	20	3	-60	S
2	-40	3	-120	S
3	72	8	576	A
4	-68	5	-340	S
5	16	5	80	A
6	20	6	-120	S
7	12	2	-24	S
8	8V	1	8	A

$I_1 - I_2 = 3A$

$I_4 = 5A = I_5$

KCL at D:

$-2 + 3 - I_8 = 0$

$\hookrightarrow I_8 = 1A$

KCL at A:

$-3 + I_3 - 5 = 0$

$\hookrightarrow I_3 = 8A$

KCL at C:

$5 - I_6 + \overset{=1}{I_8} = 0$

$\hookrightarrow I_6 = 6A$

$V_1 = 20V$, $V_3 = I_3 R = (8)(9) = 72V$

KVL for 3:

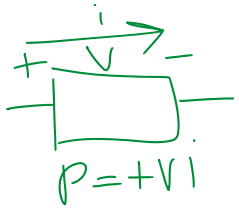
$12 - 20 + V_8 = 0 \Rightarrow V_8 = 8V$

KVL for 1:

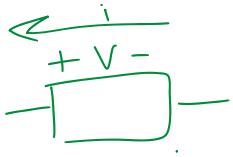
$-20 + V_2 + \overset{=72}{V_3} - 12 = 0 \Rightarrow V_2 = -40V$

KVL for 2:

$-\overset{=72}{V_3} - V_4 - 16 + 20 = 0 \Rightarrow V_4 = -68V$



$P = +Vi$



$P = -Vi$

Check $\Sigma P = 0$ ✓

Q 2: [20 marks] Using Circuit Reduction (Reducing the resistors to one resistor) and basic circuit laws (KVL, KCL, Ohms Law, current division principle and voltage division principle) for the circuit shown in Figure 2, Calculate the equivalent resistance (total resistance) as seen from the source, I_1, I_2, V and the power supplied by the 4A source.

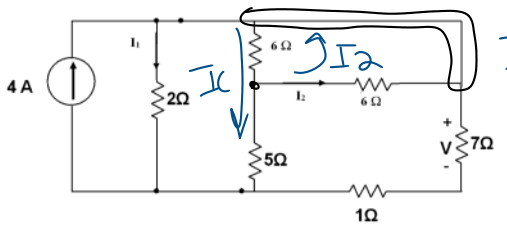
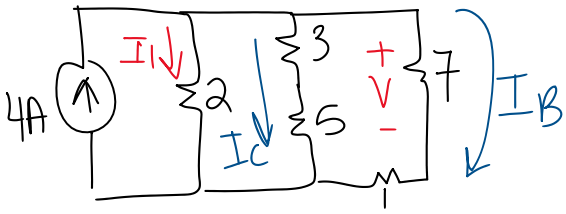


Fig.2: Circuit for Q 2

$$I_2 = -I_C\left(\frac{6}{6+6}\right) = -\frac{1}{3} = -0.33A$$

① $6 \| 6 = \frac{6\times 6}{6+6} = 3$



CD

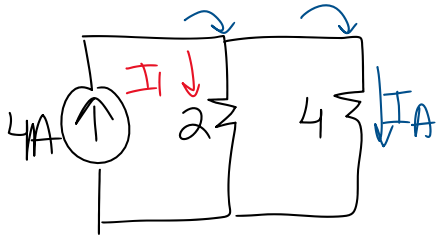
$$I_B = I_A\left(\frac{3+5}{7+1+3+5}\right) = 2/3\ A$$

$$V = 7(I_B) = 4.67V$$

$$I_C = I_A - I_B = 2/3\ A$$

② $(3+5)\|(7+1) = 8\|8$

$$= \frac{8\times 8}{8+8} = 4$$



CD

$$I_1 = 4\left(\frac{4}{2+4}\right) = \frac{8}{3} = 2.67A$$

$$I_A = 4 - I_1 = \frac{4}{3}\ A$$

$$R_{eq} = 2\|4 = \frac{2\times 4}{2+4} = 4/3\ \Omega$$

Q 3 [20 marks]: In the circuit shown in Fig. 3, Using Mesh analysis:
Identify your meshes, assign *clockwise* mesh currents, and write the mesh equations for the meshes assigned. [10]

Simplify the equations you wrote in part (A). [3]

Solve the simplified system in part (B). Provide numerical answers of your loop currents. [4]

Express V_x in terms of mesh currents. Provide numerical answer for V_x. [3]

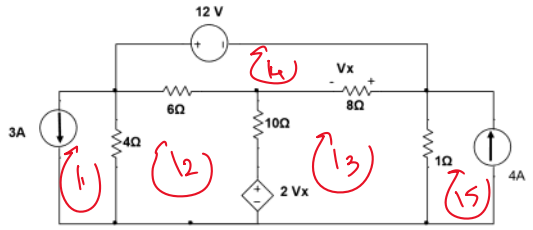


Fig. 3: Circuit for Q 3 - Mesh Analysis

$I_1 = -3A$, $I_5 = -4A$

Mesh 2

$4(I_2 - I_1) + 6(I_2 - I_4) + 10(I_2 - I_3) + 2V_x = 0$

$V_x = 8(I_4 - I_3)$

$I_2(4+6+10) + I_3(-10 - \overset{(2\times 8)}{16}) + I_4(-6+16) = 4(-3)$

$I_2(20) + I_3(-26) + I_4(10) = -12$ (1)

Mesh 3 $V_x = 8(I_4 - I_3)$

$-2V_x + 10(I_3 - I_2) + 8(I_3 - I_4) + 1(I_3 - \overset{=-4}{I_5}) = 0$

$I_2(-10) + I_3(16+10+8+1) + I_4(-24) = -4$ (2)

Mesh 4

$12 + 8(I_4 - I_3) + 6(I_4 - I_2) = 0$

$I_2(-6) + I_3(-8) + I_4(8+6) = -12$ (3)

$I_2 = -5.42A$ $I_3 = -6.32A$ $I_4 = -6.79A$

$V_x = 8(I_4 - I_3) = -3.76V$

Q 4 [20 marks]: In the circuit shown in Fig. 4, Use nodal analysis:

Write the nodal equations for the non-reference nodes. The reference node is at the bottom of the circuit. [9]

Simplify the equations you wrote in part (A). [4]

Solve the simplified system in part (B). Provide numerical answers for node voltages. [3]

Express I_x, I_y, and V_x in terms of node voltages, and provide numerical values for I_x, I_y, and V_x [4]

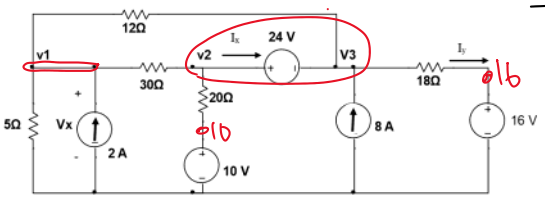


Fig. 4: Circuit for Q 4 – Nodal Analysis

Supernode

$V_2 - V_3 = 24$ (1)

$\hookrightarrow V_2 = V_3 + 24$

$$\frac{V_2 - V_1}{30} + \frac{V_2 - 10}{20} - 8 + \frac{V_3 - 16}{18} + \frac{V_3 - V_1}{12} = 0$$

$$V_1\left(-\frac{1}{30} - \frac{1}{12}\right) + V_2\left(\frac{1}{30} + \frac{1}{20}\right) + V_3\left(\frac{1}{18} + \frac{1}{12}\right) = \frac{10}{20} + 8 + \frac{16}{18}$$

$$V_1\left(-\frac{7}{60}\right) + V_2\left(\frac{1}{12}\right) + V_3\left(\frac{5}{36}\right) = \frac{169}{18} \quad (2)$$

Node 1

$$\frac{V_1 - 0}{5} - 2 + \frac{V_1 - V_2}{30} + \frac{V_1 - V_3}{12} = 0$$

$$V_1\underbrace{\left(\frac{1}{5} + \frac{1}{30} + \frac{1}{12}\right)}_{= 19/60} + V_2\left(-\frac{1}{30}\right) + V_3\left(-\frac{1}{12}\right) = 2 \quad (3)$$

$V_1 = 26.15\,V \qquad V_2 = 70.98\,V \qquad V_3 = 46.98\,V$

$V_x = V_1 - 0 = 26.15\,V$

I_x: Node 2: $\frac{V_2 - V_1}{30} + \frac{V_2 - 10}{20} + I_x = 0$

$I_y = \frac{V_3 - 16}{18}$

Q 5: [20 marks] For the circuit shown in Fig. 5,

Determine the Thevenin's Equivalent at terminals **a** and **b**. For finding V_{TH} **only use Superposition principle** (if any other circuit analysis technique is used for finding Vth, marks will not be given). For finding R_{TH} any circuit analysis method of your choice can be used. [14]

Determine the value of the load resistor (R_L) for maximum power transfer to the load. [2]

Determine the value of the maximum power transferred to the load. [4]

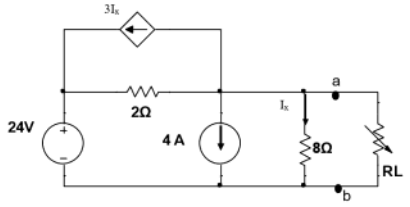
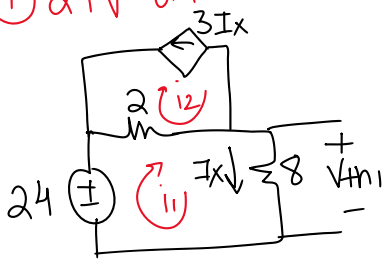


Fig. 5: Circuit for Q 5 – Thevenin's Theorem

① 24V on



$i_2 = -3I_x = -3i_1$

Mesh 1

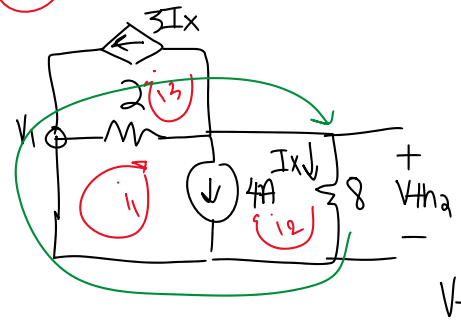
$-24 + 2(i_1 + 3i_1) + 8(i_1) = 0$

$i_1(2+6+8) = 24$

$i_1 = 1.5$

$V_{th_1} = i_1(8) = 12V$

② 4A on



$i_3 = -3I_x = -3i_2$

Supermesh

$i_1 - i_2 = 4$

$2(i_1 + 3i_2) + 8(i_2) = 0$

$i_1(2) + i_2(6+8) = 0$

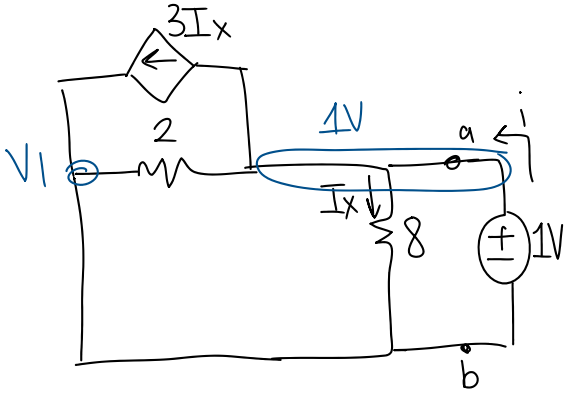
$i_1 = 3.5 \quad i_2 = -0.5$

$V_{th_2} = 8(i_2) = -4V$

$V_{th} = V_{th1} + V_{th2} = 12 - 4 = \underline{\underline{8V}}$

R_th:

$R_{th} = 1V/i$



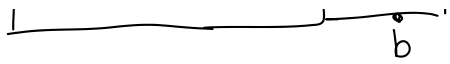
$I_x = \frac{1}{8}$

Node 1

$\frac{V_1 - 1}{2} - 3\left(\frac{1}{8}\right) = 0$

$\hookrightarrow V_1\left(\frac{1}{2}\right) = \frac{1}{2} + \frac{3}{8}$

$V_1 = 1.75$



$\rightarrow$ "(2) a 8

$V_1 = 1.75$

Node 2

$$\frac{1-1.75}{2} + \frac{3}{8} + \frac{1}{8} - i = 0$$

$$i = 1/8^{A}$$

$$R_{th} = 1^{V} / 1/8^{A} = \underline{\underline{8\,\Omega}}$$

$$R_L = R_{th} = 8\,\Omega$$

$$P_{max} = \frac{V_{th}^{2}}{4R_{th}}$$